DIGOXIN

Narrow therapeutic index
Check pulse before dose

CHEMICAL NAME 3β-[(O-2,6-dideoxy-β-D-ribo-hexopyranosyl-(1→4)-O-2,6-dideoxy-β-D-ribo-hexopyranosyl-(1→4)-2,6-dideoxy-β-D-ribo-hexopyranosyl)oxy]-12β,14-dihydroxy-5β-card-20(22)-enolide

CLASS Cardiac glycoside (positive inotrope, AV node slowing)

TRADE NAMES Lanoxin · Lanoxin PG · Digitek

THERAPEUTIC USES / INDICATIONS

- Atrial fibrillation/flutter; ventricular rate control (especially sedentary or heart-failure patients).
- Heart failure with reduced EF (symptom control, ↓ hospitalisation) in sinus rhythm or AF.

ADMINISTRATION

- **PO** once daily at the same time (tablets, elixir — measure accurately).
- **IV** slowly over 10–20 min (loading) — not IM.
- Check apical pulse for 1 full minute before each dose.

DOSAGES

- **Rapid loading (AF):** 0.75–1.5 mg in divided doses over 24 h.
- **Maintenance:** 62.5–250 mcg once daily (by renal function, age, weight).
- **Heart failure:** 62.5–125 mcg daily.
- **Children:** weight- and age-based (specialist).
- Target level 0.5–1 ng/mL in HF (up to 2 ng/mL AF); take level ≥6 h post-dose.

PHARMACOKINETICS

- **Onset:** IV 5–30 min; PO 30 min–2 h. **Peak:** 1.5–6 h.
- Large Vd (binds skeletal muscle) — dose on lean weight.
- Excreted mainly unchanged in urine. **t½:** 36–48 h (up to 5 days in renal failure).

PHARMACODYNAMICS

- Inhibits **Na^+/K^+-ATPase** in myocardium → ↑ intracellular Ca^{2+} → ↑ contractility (positive inotrope).
- ↑ vagal tone → slows SA node and AV node conduction (negative chronotrope).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, anorexia (early toxicity sign).
- Fatigue, dizziness.
- Gynaecomastia (long-term).

ADVERSE EFFECTS (SERIOUS)

- Digoxin toxicity: nausea, vomiting, diarrhoea, confusion, yellow-green vision (xanthopsia), bradycardia, heart block, any arrhythmia (classic: atrial tachycardia with block, bidirectional VT).
- Toxicity ↑ with hypokalaemia, hypomagnesaemia, hypercalcaemia, renal impairment, elderly.
- Antidote: digoxin-specific antibody fragments (DigiFab).

CONTRAINDICATIONS & CAUTIONS

- 2nd/3rd-degree heart block (without pacemaker); intermittent complete block.
- WPW syndrome with AF (↑ accessory pathway conduction).
- Ventricular tachycardia/fibrillation; HOCM (unless AF).
- Constrictive pericarditis.

KEY INTERACTIONS

- **Amiodarone, verapamil, diltiazem, clarithromycin, quinidine, spironolactone, ciclosporin:** ↑ digoxin levels.
- **Loop/thiazide diuretics, steroids, amphotericin:** hypokalaemia → toxicity.
- **β-blockers:** additive bradycardia.
- **Antacids, cholestyramine:** ↓ absorption.

NURSING MANAGEMENT

- Hold dose and report if apical pulse <60/min (adult; <70 child, <90 infant) or new irregularity.
- Monitor K^+, Mg^{2+}, creatinine, digoxin levels.
- Ask about nausea, visual changes, confusion.
- Daily weights, fluid balance in HF.
- Check ECG for heart block, arrhythmias.

PATIENT EDUCATION

- Learn to take your pulse; do not take the dose if below 60 and contact your doctor.
- Report nausea, vomiting, blurred or yellow vision, confusion.
- Do not change brands or double doses.
- Eat potassium-rich foods if on diuretics.

KEY POINTS

- Na^+/K^+-ATPase inhibitor — inotrope + AV node slowing.
- Hypokalaemia → toxicity; xanthopsia.
- Narrow therapeutic index; long t½.
- Hold if pulse <60; antidote DigiFab.